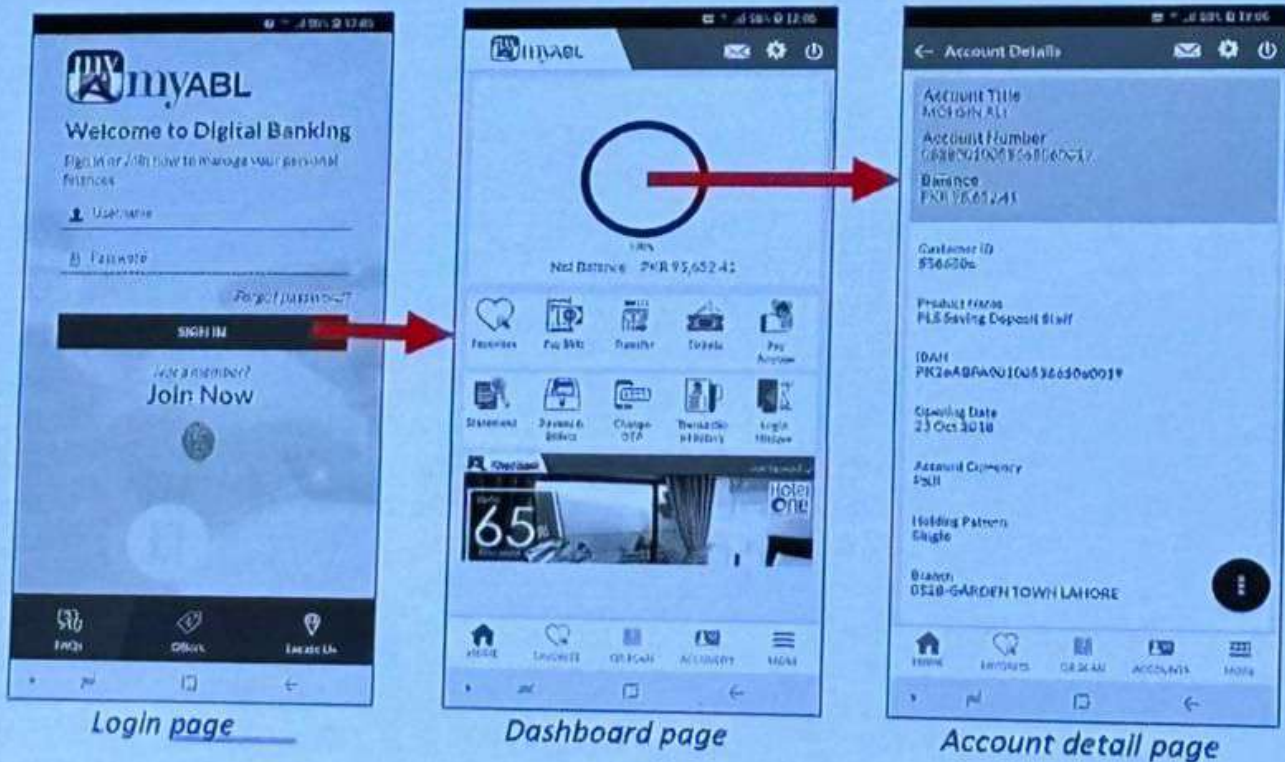


- | | | | |
|-----|--------------------------------|----|------------------------|
| iv. | Consistency and standards | | recover from errors |
| v. | Error prevention | x. | Help and documentation |
| vi. | Recognition rather than recall | | |

Assume you are preparing a new interface for an online banking application. The major problem the bank is facing is the **memorability** and **learning** of the software, i.e. users have always complained that they forget how to use this application, despite being frequent users. Usability experts believe that the major problem is in the perception of the system to users. Many users have complained that they have to learn every new process and interaction, rather than performing the actual banking transaction, such as paying a utility bill or a transaction for funds transfer.

As an expert, the bank has assigned you the task to improve the usability of their application. The following are the screenshots of the existing login page, dashboard, and account details page:



- a) You suspect that different users use the account detail page in several quite distinct ways. Instead of simply checking balance, they wish to use this page to do further transactions, such as payment of bills or online cash transfer, etc. Based on this reading, you feel that some design change of the dashboard page should be conducted to remove this problem. Secondly, you also want to incorporate the main aspects of social mechanisms of "coordination among pages" and "awareness of the users where they are and what they intend to do?".

What will be the basic design changes that you will incorporate on the dashboard page? State any two major changes. (2 marks)

- b) A study of the system data logs shows two surprising patterns of behaviour. Some users repeatedly enter customer ID and Branch ID to use for transactions instead of their account number. The other pattern includes long pauses between entering each piece of data, with the same users accessing help and documentation pages in alternation with data entry pages, i.e. the users try to access help when completing a transaction. When they access help in the middle of the transaction process, the application takes them to the dashboard page after completion of the 'help' process.
- I. How will you handle these two patterns in your new design? (2 marks)
 - II. What usability analysis procedures might you carry out to improve the usability of the system for these two classes of the user? (2 marks)
 - III. What improvements might be experienced by each class as a result? (2 marks)
- c) After the project has begun, it is decided that the new design should also support two further classes of users: those who do not want to make a transaction online, but simply need to find out through their application, what services their branch can offer. And the second class is of the users who want to make telephonic calls from their application to the call-center to find out more about the services their physical

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- How will you handle these two patterns in your new design? (2 marks)
- What usability analysis procedures might you carry out to improve the usability of the system for these two classes of the user? (2 marks)
- What improvements might be experienced by each class as a result? (2 marks)

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(2 marks)

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What design changes you will use to customize the user interface for these classes of users, and what improvements might they experience as a result? (2 marks)

Q2: The development manager of a website for online book-buying has asked you to carry out a heuristic evaluation of its usability. He has specifically proposed the three heuristics listed below.

1. "There should be between five and nine navigation options on each page."

2. "There should be a good match between the navigation buttons and the users' goals."

IBA

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1. "There should be between five and nine navigation options on each page."
2. "There should be a good match between the navigation buttons and the users' goals."
3. "It should be easy for users to change their plans."

- (a) Has the manager misunderstood heuristic evaluation? Briefly justify your answer. (2 marks)
- (b) Please comment on the above three heuristics suggested by the manager. For each of the proposed heuristics, your comments should include:

- i. Any theoretical justification for (or against) this heuristic;
- ii. Any additional evaluation steps that might be required in applying it; and
- iii. The likely impact of such evaluation on the system design. (3 marks)

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Q3: (a) How would you differentiate between principles, standards, guidelines, and heuristics, using at least one example of each to illustrate? (2 marks).

(b) You have studied the following different evaluation methods:

- iv. Heuristic Evaluation
- v. Cognitive Walkthrough
- vi. Think a loud
- vii. Usability Testing

In your opinion, which type of method is best used to evaluate different user groups and in what situations? Please add your justifications for selecting a method for a particular user group. The User Group is divided into elders, teenagers, disabled people, and illiterates. (3 marks)

END OF PAPER

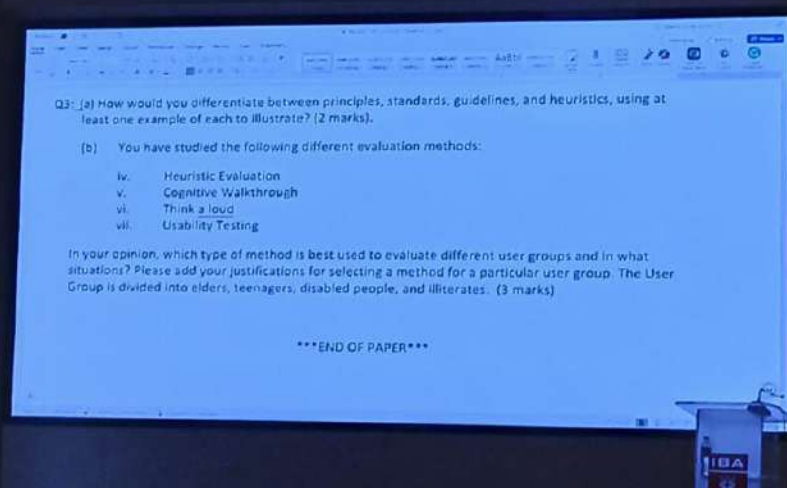
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